

[Logic Flow]



[Smart Agriculture Assistant, 4]

[An AI-powered agriculture assistant that combines soil, weather and historical crop data to help farmers make smarter decisions on crop selection, irrigation and fertilizer usage.]

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Farmers often make critical decisions using fragmented information about soil, weather, crops and past yields.

Changing environmental conditions make it difficult to determine:

Which crop is suitable?

When irrigation is required?

How much fertilizer is appropriate?

How historical performance can guide current decisions?

Fragmented agricultural data — Soil, weather and historical crop information are often available separately, making combined analysis difficult.

Changing environmental conditions— Rainfall, temperature and soil moisture can change rapidly, affecting irrigation and crop suitability.

Limited decision support— Farmers need actionable recommendations rather than raw agricultural data.

The challenge is not lack of data — it is turning agricultural data into timely, understandable decisions.

Smart Agriculture Assistant helps farmers turn agricultural data into actionable farm recommendations.

01

Farm & Environmental Data

Users provide:

Nitrogen (N)
Phosphorus (P)
Potassium (K)
Soil pH
Rainfall
Soil moisture
Temperature

02

AI + Data Analysis

AgriSense combines:

MongoDB agricultural data
PostgreSQL farm information
Historical crop performance
Weather and soil information
RAG-based agricultural knowledge

It then analyzes the conditions to generate recommendations.

03

Actionable Farm Decisions

The system provides:

Recommended Crop
Irrigation Recommendation
Fertilizer Recommendation
Historical Crop Insights
AI Agriculture Chatbot

“A farmer can enter current farm conditions and receive understandable recommendations for the next farming decision.”

Technology Stack

- **Frontend:** Responsive web dashboard for farm analysis, historical insights and AI chat.
- **Backend:** Python, FastAPI, Rest APIs
Handles recommendations, historical insights and chatbot requests.
- **Database:** MongoDB, PostgreSQL, PyMongo, MongoDB Aggression
MongoDB → dynamic agricultural/environmental data
PostgreSQL → structured farm/master data
Aggregation → crop performance analysis
- **AI/ML:** Rag-based Knowledge Retrieval, recommendation engine, AI ChatBot. Agricultural knowledge is retrieved from MongoDB to generate relevant responses.
- **Infrastructure:** Local Development, VS Code, Python Virtual Environment
- **Other tools:** Mongosh, MongoDB Atlas Charts, Github

Key Features / USPs

- **AI Agriculture Assistant**
Provides recommendations for **crop selection, irrigation and fertilizer usage** based on farm conditions.
- **RAG Knowledge Retrieval**
Retrieves relevant agricultural knowledge from MongoDB before answering user questions.
- **Data-Driven Insights**
MongoDB aggregation analyzes **historical yield, rainfall, fertilizer usage and production**.
- **Polyglot Data Architecture**
PostgreSQL + MongoDB allows structured farm information and flexible agricultural data to work together.

[HOW IT WORKS]

[PART 1]

Enter Farm Conditions

The farmer enters soil and environmental parameters through the AgriSense dashboard.

[PART 2]

AI Recommendation Engine

The system analyzes soil, rainfall, moisture and temperature conditions to generate farm recommendations.

[PART 3]

MongoDB + PostgreSQL + Knowledge Base

The system uses:

- Soil data
- Weather data
- Historical crop data
- Agricultural knowledge
- Farm information

[PART 4]

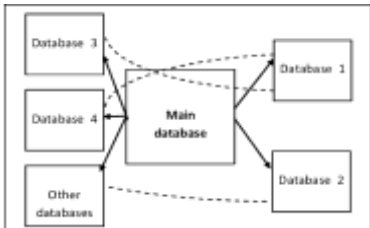
Actionable Agriculture Insights

The farmer receives:

- Crop recommendation
- Irrigation advice
- Fertilizer guidance
- Historical insights
- AI chatbot answers

[Flow: Farm Input → Data Processing → AI + RAG Analysis → Recommendation → Farmer Decision]

[DATA FLOW]



[01 · INPUT]

Farm & Environmental Data:

N
P
K
pH
Rainfall
Soil Moisture
Temperature



[02 ·
PROCESSING]

**Data
Analysis & AI:**

Validate farm inputs
Analyze environmental conditions
Apply recommendations rules
Analyze historical crop performance
Retrieve relevant agricultural knowledge



[03 ·
STORAGE]

**Dual Database
Architecture:**

PostgreSQL
→ Farm/master information
MongoDB
→ Soil, weather, historical data, recommendation s & knowledge base



[04 ·
RETRIEVAL]

**Intelligent
Data Retrieval:**

The system retrieves:
Relevant historical records
Crop performance patterns
Agricultural knowledge
Environmental information
RAG retrieves knowledge relevant to the farmer's question.



[05 · OUTPUT]

**Actionable
Results:**

Crop Recommendation
Irrigation Recommendation
Fertilizer Recommendation
Historical Insights
AI Chatbot Response